DASE7506 MP1 Report

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1 Introduction

The submitted serial two-student predictor scores 1.49693237 bits per byte (BPB) on the complete test split, compared with 2.10126027 for the original course baseline. Its local median CPU scoring-time ratio is 4.646. This report explains the training procedure and the evidence behind those results, including equal-target distillation controls, an inference-gate ablation, and training and evaluation costs. Resource acceptance on the course reference machine remains unverified.

The assignment uses the supplied WikiText-2 training text, fixed BPE-2048 tokenizer, and fixed evaluator. CPU scoring must take at most five times the baseline time, use at most 4 GiB of peak RAM, and require at most 64 MiB of uncompressed inference assets. The method uses a three-member teacher only during training and retains two smaller students for inference. A causal confidence gate runs the second student conditionally.

2 Task and evaluation protocol

The course provides fixed train, validation, and test splits, a train-fitted BPE-2048 tokenizer, a four-block width-128 GPT baseline, and a scorer. The scorer evaluates independent causal windows of at most 256 targets. It sums next-token negative log-likelihood and divides by the raw split's UTF-8 byte count and ln(2) to obtain BPB. Lower BPB is better. The complete test split has 428,405 scored targets and 1,292,013 raw bytes. The report uses complete-validation scores for development comparisons and complete-test scores only for explicitly identified frozen predictors.

The original 1,200-step baseline scores 2.10126027 test BPB under CPU FP32 evaluation. The submitted predictor lowers this by 0.60432790 BPB, a relative reduction of 28.76%. That overall difference combines a larger model, longer training, teacher targets, snapshot averaging, two students, and temperature calibration. It cannot be attributed to any one of those components. The matched controls below answer narrower questions: whether teacher targets help a specified student recipe at equal training targets, and how the gate changes validation quality for fixed student checkpoints.

3 Method

3.1 Frozen teacher and student training

The teacher is an equal-probability mixture of three width-192, four-block GPTs whose component weights were trained only on the supplied training text. One component has no dropout and two use dropout probability 0.05. Its shared logit temperature, 1.05, was selected on validation. The teacher's complete-validation BPB is 1.508368. It is frozen before student training, supplies a distribution for each sampled training prefix, and is never loaded by the submitted inference model [1].

Each selected student has four transformer blocks and width 256. Seeds 17 and 23 each run 14,400 updates with batch size 32 and sequence length 256, processing 117,964,800 training targets per run. The student objective places weight 0.25 on the hard target cross-entropy and weight 0.75 on cross-entropy against the frozen teacher distribution. Both students start from random initialization. The same training split supplies prefixes and hard targets; no validation or test targets enter the student updates.

For each student, seven checkpoints from steps 12,600 through 14,400, spaced every 300 steps, are combined into one exponential parameter average with decay 0.7. This late average adds no inference model. The two averaged students have equal probability-mixture weights. Validation selects a separate logit temperature for each student; both selected values are 0.975. The training objective, snapshot schedule, and student temperatures are distinct choices: the paired distillation controls hold the architecture and processed targets fixed, while the later inference ablations hold student checkpoints fixed.

3.2 Causal confidence gate

Student 1 is evaluated on every independent scoring window. The gate reads the largest calibrated next-token probability at the window's first observed position. If that maximum is below 0.16, student 2 runs for the entire window and the output at every position is the equal mixture of both calibrated student distributions. Otherwise, the output is student 1 alone. The first-position decision depends on the observed prefix, not on the target to be scored or any future position. No state is carried between windows or calls. The scorer uses four PyTorch intra-op threads and no additional Python worker threads.

The gate is a cost control rather than an accuracy mechanism. Skipping student 2 can worsen the prediction at positions later in a confident window because one first-position decision governs the whole window. This coarse decision avoids a second forward pass for selected windows and keeps the evaluation path simple. A more fine-grained gate might change the quality and latency frontier, but it is not part of the submitted system and is not evaluated here.

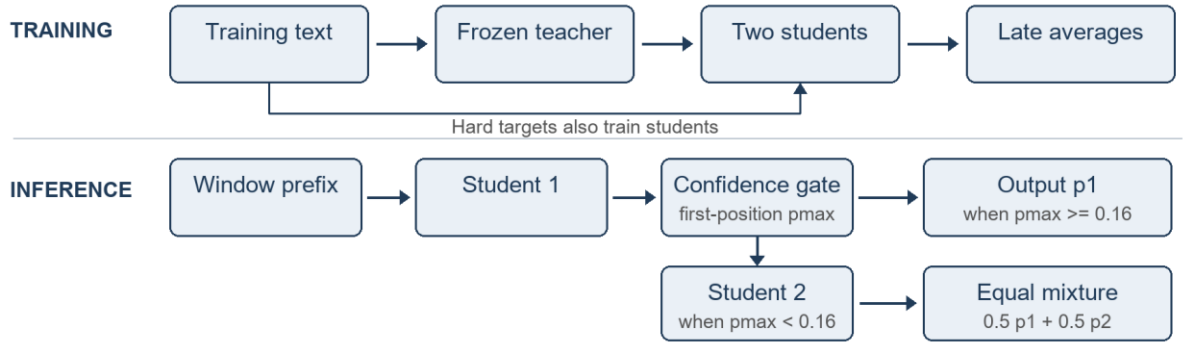


Figure 1 Training and inference paths.

4 Experimental design and development

4.1 Controls and measurement boundaries

All development selections use the fixed validation split. Within each stated training-budget pair, runs use the same architecture, seed, batch size, update count, and number of processed targets. Such a comparison controls data exposure, but teacher distillation adds teacher forward computation and therefore does not match training time. Cross-budget comparisons are described separately because learning-rate schedules, checkpoint selection, and averaging can change together. Unless otherwise noted, reported validation BPB comes from complete-split CPU FP32 scoring.

4.2 Early architectural screens

A same-seed, 1,200-update normalization screen asked whether RMSNorm should replace the supplied LayerNorm.[2] At equal targets, seed-17 RMSNorm scored 2.07796980 validation BPB against 2.07108351 for LayerNorm, a deterioration of 0.00688629 BPB with nearly identical scoring

time. RMSNorm was not promoted, and a second seed was not run for that screen. This negative result matters because it prevents the final improvement from being attributed to a normalization change that the final students do not use.

A feed-forward screen initially favored GEGLU.[3] At 2,400 updates it scored 1.82667069 against GELU's 1.88050155 for seed 17, and 1.83023854 against 1.87531450 for seed 23. Repeating the matched comparison at 4,800 updates changed the interpretation: seed 17 favored GEGLU by only 0.00250917 BPB, while seed 23 favored GELU by 0.00139761 BPB. The longer-budget evidence did not support a stable GEGLU advantage, so the main line retained GELU. Both comparisons are validation results, and neither mechanism appears in the submitted predictor as an added GEGLU branch.

At 4,800 updates with width 128, raising peak learning rate from 0.001 to 0.0013 improved validation BPB in seeds 17 and 23 from 1.75244 and 1.75056 to 1.73417 and 1.73327. Under the revised schedule and matched targets, increasing width to 160 yielded 1.68941 and 1.69231. A separate 9,600-update matched width comparison again favored width 160 over width 128. This was evidence for a capacity benefit under those recipes, accompanied by more parameters and compute; it did not isolate width as the cause of the final system's overall gain.

4.3 Stable student recipe

A width-320 hard-label study did not yield a valid control. Peak learning rates 0.001 and 0.0007 produced non-finite training values. Runs at 0.0003 and 0.0001 were stopped after loss collapse. The four attempts ended after 1,800, 600, 4,600, and 2,700 logged steps respectively. No successful width-320 result supports a ranking against width 256; selecting width 256 reflects a stable tested recipe rather than a global width optimum.

The first width-256 teacher-versus-hard-label study used seed 17 for 9,600 updates, or 78,643,200 targets. The hard-label control scored 1.669731 validation BPB and soft weight 0.50 scored 1.523223, a reduction of 0.146509. This paired result motivated a 14,400-update study with seven-snapshot averaging and a wider soft-weight sweep. The separate 14,400-step hard-label control scored 1.726626, worse than the 9,600-step control, but the schedule and checkpoint-selection procedure changed. That cross-budget observation does not establish that additional training generally harms performance.

5 Matched controls and mechanism ablations

5.1 Equal-target distillation comparisons

Updates	Seed	Soft weight	Targets (M)	Validation BPB	Train (s)
9,600	17	0.00 hard CE	78.643	1.669731	173.918
9,600	17	0.50	78.643	1.523223	352.129
14,400	17	0.00 hard CE	117.965	1.726626	264.515
14,400	17	0.25	117.965	1.582153	531.628
14,400	17	0.50	117.965	1.515954	531.066
14,400	17	0.75	117.965	1.497079	531.434

Table 1 Complete-validation CPU FP32 BPB and measured student training time.

At 14,400 updates, increasing soft weight from zero to 0.75 lowers seed-17 validation BPB by 0.229546. The corresponding student training time rises from 264.515 to 531.434 GPU seconds, about 2.01 times as long before teacher training and search are counted. This is a quality gain at equal target exposure, not a free gain at equal compute. At 9,600 updates, the weight-0.50 student similarly improves by 0.146509 BPB while its measured training time is about twice that of hard-label training.

A later seed-17 weight screen tested 0.65, 0.75, and 0.85, obtaining 1.500808, 1.497079, and 1.502125 validation BPB. Seed 23 independently favored 0.75 over 0.50, scoring 1.497176 versus 1.513088. The two selected weight-0.75 students average 1.497128 BPB as separate single models. These results support the chosen distillation procedure under the tested recipe. Only seed 17 has a 14,400-step zero-distillation control, so the evidence is stronger for the selected recipe than for a universal effect across seeds or model widths.

5.2 Ensemble, gate, and calibration

Configuration	Validation BPB	Inference behavior
Student 1, seed 17 average	1.497079	One student
Student 2, seed 23 average	1.497176	One student
Equal mixture, no gate	1.478205	Both students every window
Gate 0.16, temperature 1.0	1.481876	Student 2 on selected windows
Gate 0.16, temperature 0.975	1.480946	Final frozen predictor

Table 2 Complete-validation mechanism ablations.

The ungated equal mixture improves on either uncalibrated single student by about 0.019 BPB. Applying the 0.16 gate at temperature 1.0 adds 0.003671 validation BPB relative to that mixture. Calibrating both student temperatures to 0.975 recovers 0.000929 BPB relative to the otherwise identical temperature-1.0 gate. The final quality cost is 0.002742 BPB versus the uncalibrated always-on mixture. These are complete-validation comparisons, not full-test tuning results.

The validation record also contains an alternative gate threshold of 0.25 with 1.479932 BPB, compared with 1.481876 for the selected uncalibrated threshold 0.16. The less selective threshold improves quality but can call student 2 on more windows. The staged records do not provide a complete, same-session latency table or the fraction of windows invoking student 2 for every threshold. Consequently, this report does not claim that the threshold sweep establishes an exact calibrated speed-quality frontier. The local full-test timing records below support the resource status of each frozen version, with the limitations of separate measurement sessions.

6 Frozen candidates and resource measurements

6.1 Disclosed evaluation sequence

Table 3 compares inference configurations using the complete validation split only. All configurations use the same two fixed student averages. The always-on mixture achieved the lowest validation BPB but had the highest CPU cost and limited headroom under the 5× evaluation-time budget. A causal gate with threshold 0.16 was therefore evaluated as a validation-stage quality–cost trade-off. After the weights, gate threshold, temperatures, and implementation were fixed, the final configuration was recorded in the freeze manifest and evaluated on the test split.

Predictor	Validation BPB	Scoring-time ratio vs. baseline
Always-on	1.478205	4.934
Gate 0.16	1.481876	4.714

Table 3 Validation-only comparison used to select the inference configuration before final test evaluation.

6.2 Final score and measured budget

The checkpoint's full SHA-256 appears in the submitted freeze manifest. Three fresh-process complete-test CPU FP32 runs produced the same 1.4969323726595785 BPB. Candidate and baseline processes were run alternately. Scorer seconds exclude startup, while peak process RSS includes imports, loading, and scoring. The recorded scorer and tokenizer hashes match the freeze manifest. These local checks do not establish reference-machine compliance.

Run	Final (s)	Baseline (s)	Paired ratio
1	24.280	5.057	4.801x
2	24.195	5.234	4.622x
3	23.823	5.208	4.574x

Table 4 Final complete-test CPU FP32 scorer seconds.

Resource	Final predictor	Course limit	Local status
CPU scoring time	4.646x baseline median	At most 5x	Pass
Peak process RSS	1.618 GiB	At most 4 GiB	Pass
Inference assets	28.772 MiB	At most 64 MiB	Pass

Table 5 Local resource accounting for the same final frozen predictor.

The final median timing ratio has 0.354x headroom to the local 5x boundary; the slowest paired ratio has 0.199x headroom. That is a narrow timing margin, so a different reference CPU, software stack, or thermal state could change the conclusion. Peak memory and asset size have much larger measured margins.

7 Training cost, limitations, and reproducibility

7.1 Training and search cost

Training scope	Device	Processed targets	Training seconds
Eight width-256 student runs	GPU	943,718,400	3,979.42
All 56 completed GPU runs	GPU	3,342,664,704	7,820.15
All 4 completed CPU runs	CPU	29,573,120	1,128.70

Table 6 Completed-trajectory lower bounds from the staged training ledger.

The final two distilled student runs alone processed 235,929,600 targets in 1,062.99GPU seconds. Across eight recorded width-256 student runs, the total reaches 943,718,400 targets and 3,979.42 GPU seconds. Teacher training and teacher selection are additional costs, and the 56 completed GPU runs include broader architectural search. The ledger excludes stopped width-320 attempts without complete metrics, validation, checkpoint loading, CPU scoring, and other search overhead. It therefore reports lower bounds, not an end-to-end project wall time.

7.2 Limits of the evidence

The strongest training-objective evidence is the same-seed, same-target hard-label-versus-distillation pair. Its inference architecture is held constant, but the teacher makes training about twice as expensive. Only one seed has a 14,400-step zero-distillation control. Snapshot averaging, learning-rate schedule, and training duration change together between the 9,600- and 14,400-step studies, so that cross-budget comparison cannot isolate any one of those choices. The normalization and GEGLU screens concern paths that were not promoted to the final predictor.

The gate's quality cost is measured directly on validation with fixed student checkpoints. The resource figures come from separate local test-time sessions, and the staged records do not contain a full same-session calibrated gate-versus-ungated latency sweep or per-threshold gate-use fractions. The selected threshold can therefore be described as a validation-supported local operating point, not as a proven optimum. The gate uses the first position to decide the entire window; a corpus with different confidence patterns could move the quality and speed frontier. Reference-machine timing remains unmeasured.

8 Conclusion

The final serial confidence-gated pair achieves 1.49693237 full-test BPB on the supplied benchmark, compared with 2.10126027 for the original baseline. Equal-target validation controls support distillation under the selected student recipe, while validation ablations show that the gate sacrifices 0.002742 BPB relative to the ungated mixture. Its measured local resource profile falls within the stated limits, with limited CPU-time headroom.

References

[1] G. Hinton, O. Vinyals, and J. Dean. Distilling the Knowledge in a Neural Network. arXiv:1503.02531, 2015.

[2] B. Zhang and R. Sennrich, “Root Mean Square Layer Normalization,” Advances in Neural Information Processing Systems, vol. 32, 2019.

[3] N. Shazeer, “GLU Variants Improve Transformer,” arXiv preprint arXiv:2002.05202, 2020.